CAP: Expanding AI Research, Education, and Workforce Development Capacity for Equity and Innovation at Texas State University

Solicitation: NSF 23-506 — Expanding AI Innovation through Capacity Building and Partnerships (ExpandAI)
Program Track: Capacity Building Pilot (CAP)
Principal Investigator (PI): Dr. Amina Lopez, Associate Professor, Department of Computer Science, Texas State University
Co-Principal Investigator (Co-PI): Dr. Ethan Wu, Assistant Professor, Department of Curriculum and Instruction, Texas State University
Duration: 24 months (2025–2027)
Requested Budget: $398,700
Institution Type: Hispanic-Serving Institution (HSI)

PROJECT SUMMARY

Overview: Texas State University proposes a two-year capacity-building initiative under NSF's ExpandAI program to establish a sustainable ecosystem for Artificial Intelligence (AI) education, research, and workforce development. The project focuses on leveraging AI for Education, Equity, and Workforce Development by empowering faculty and students—particularly from Hispanic, first-generation, and low-income backgrounds—to engage in ethical, inclusive, and interdisciplinary AI innovation.

Intellectual Merit: This project will contribute new insights into effective AI-integrated pedagogy and research capacity building at Minority-Serving Institutions. It will advance the understanding of how diverse learners acquire computational and AI literacy through interdisciplinary collaboration and applied research. Faculty development workshops and student research experiences will generate scalable models for integrating AI into non-computing disciplines, bridging gaps between technical and social sciences.

Broader Impacts: The initiative will expand participation of underrepresented populations in AI, support the development of an inclusive AI workforce pipeline, and produce open-access curriculum and community engagement models for replication at other MSIs. By combining education, research, and industry engagement, the project aligns with NSF's vision to ensure equitable access to emerging technologies and to strengthen national innovation capacity.

PROJECT DESCRIPTION

A. Need and Potential

Texas State University, a Hispanic-Serving Institution, enrolls over 38,000 students, 44% of whom identify as Hispanic or Latino/a and nearly 50% as first-generation college students. Despite its strong Computer Science and Education programs, the institution currently lacks dedicated research clusters, training programs, and cross-disciplinary structures for AI. This proposal responds directly to NSF's call to expand participation in AI by establishing faculty and student capacity, research infrastructure, and external partnerships.
Currently, AI-related instruction is limited to upper-level computing courses. Students from education, social science, and business fields have minimal exposure to applied AI concepts. Faculty surveys indicate a growing interest in using AI tools for instruction and research but a lack of access to training, computing resources, and external partnerships. The CAP project will fill this institutional gap by launching a coordinated effort to develop curriculum, research capacity, and collaboration opportunities across disciplines.

B. Capacity Building Plan

The project will follow a structured three-phase plan:
Phase 1: Faculty Development and Infrastructure (Months 1–6) — Train 12 faculty members through the AI Faculty Fellows program, co-led by Computer Science and Education departments. Faculty will complete intensive workshops on AI fundamentals, data analytics, and machine learning ethics. Texas State will establish a small AI Learning Lab equipped with GPUs and cloud computing access through CloudBank. Each faculty fellow will design an AI module or project to integrate into existing courses.
Phase 2: Curriculum and Research Integration (Months 7–16) — Introduce three new interdisciplinary AI courses: (1) 'AI for Education and Society,' (2) 'Machine Learning Fundamentals,' and (3) 'Responsible AI and Policy.' Courses will include project-based learning and community-focused data challenges. Faculty and student teams will initiate AI research projects addressing educational equity, such as predictive analytics for academic success and natural language tools for multilingual learning support.
Phase 3: Student Engagement and Workforce Pathways (Months 17–24) — Launch the AI Scholars Program, recruiting 25 undergraduates annually, prioritizing students from underrepresented groups. Students will receive stipends, mentorship, and placements in summer internships with regional partners. A capstone AI Showcase will disseminate outcomes to the broader community, featuring student-led demos and outreach to local K–12 institutions.

Evaluation and Metrics

An external evaluator will track outcomes through qualitative and quantitative methods. Metrics will include: number of trained faculty and new AI courses; student participation demographics; research output; and partnership activities. Annual reports will measure progress toward sustainability, diversity impact, and workforce outcomes.

C. Broader Impacts

This project will expand pathways for Hispanic and first-generation students into AI careers. The faculty training and curriculum infrastructure will persist beyond the grant period, supporting sustained growth in AI instruction and research. Deliverables include: 3 new AI courses, 2 annual workshops, an open-access curriculum repository, and an active network connecting Texas State with AI Institutes and regional industry. Dissemination will occur through NSF-sponsored conferences, peer-reviewed publications, and collaborations with the AI Institute for Inclusive Learning.

D. Personnel

Dr. Amina Lopez (PI) will lead project coordination, faculty development, and AI research supervision. Dr. Ethan Wu (Co-PI) will direct curriculum design and mentoring activities. A postdoctoral associate will assist with AI research facilitation, and an advisory board—including representatives from an NSF AI Institute and local technology firms—will oversee progress and provide guidance on scaling activities.

REFERENCES CITED

1. National Science Foundation (2022). Building the Next Generation of AI Talent. 2. Anderson, M., & Garcia, L. (2021). Broadening Participation in Computing through Inclusive Pedagogy. 3. U.S. Department of Education (2023). STEM Education Strategic Plan. 4. Williams, J. & Lee, C. (2021). Integrating AI Education into Undergraduate Curricula. Journal of Computing in Higher Education. 5. AI Institutes Annual Report (2024). Expanding Access to Ethical AI Research.

BIOGRAPHICAL SKETCHES

Dr. Amina Lopez — Associate Professor, Department of Computer Science, Texas State University. Ph.D. in Computer Science (University of Texas at Austin). Research interests: machine learning, AI education, and fairness in algorithms. Publications: over 30 peer-reviewed papers in AI and data science education. Synergistic Activities: Faculty mentor for Women in Computing, Co-chair of Texas State's AI Task Force.

Dr. Ethan Wu — Assistant Professor, Department of Curriculum and Instruction. Ed.D. in Learning Technologies (University of Minnesota). Research interests: AI literacy, human-computer interaction, and educational data mining. Publications: 15 peer-reviewed journal articles. Synergistic Activities: PI of NSF-funded Data-Driven Pedagogy pilot.

BUDGET AND BUDGET JUSTIFICATION

Total Request: $398,700 over two years.
Year 1: $198,000 — Personnel ($85,000), Student Support ($40,000), Workshops/Training ($30,000), Equipment ($25,000), Travel/Dissemination ($10,000), Evaluation ($8,000).
Year 2: $200,700 — Personnel ($90,000), Student Support ($45,000), Research Supplies ($20,000), AI Showcase and Outreach ($15,000), Evaluation ($10,000), Indirect Costs ($20,700).
Budget justification emphasizes direct investment in people, training, and sustainable resources.
No voluntary cost sharing is included.

CURRENT AND PENDING SUPPORT

Dr. Lopez: No active NSF funding; prior institutional support for machine learning curriculum redesign. Dr. Wu: Current Co-PI on an internal grant for AI literacy ($12,000). Neither holds overlapping support relevant to this proposal.

FACILITIES, EQUIPMENT, AND OTHER RESOURCES

Texas State University provides dedicated computer science laboratories, high-speed computing infrastructure, and access to the Texas Advanced Computing Center (TACC). The project will establish an AI Learning Lab with 10 GPU-enabled workstations. Cloud computing services will be obtained via CloudBank. Existing classrooms, software licenses, and the university's virtual learning environment will support curriculum integration.

DATA MANAGEMENT AND SHARING PLAN

Data will be collected from research and educational activities, including student learning analytics and project outcomes. All personal identifiers will be removed to comply with FERPA and NSF policies. Anonymized datasets and curricular materials will be archived in the university's institutional repository and shared under a Creative Commons CC-BY 4.0 license. The team will document metadata standards, retention schedules, and access controls.

MENTORING PLAN

Undergraduate AI Scholars will be paired with faculty mentors for one-year research apprenticeships. Mentoring will involve biweekly progress meetings, technical skill-building sessions, and professional development in scientific communication. Graduate assistants and postdoctoral mentors will undergo mentor training workshops offered by the Office of Research and Sponsored Programs. Success will be tracked through mentor-mentee evaluations, publication co-authorships, and graduate school enrollment outcomes.

SUPPLEMENTARY DOCUMENTS

Institutional Need and Support Statement: The Office of the Provost endorses this initiative, affirming Texas State University's commitment to diversifying AI education. The university pledges matching in-kind support through computing infrastructure, faculty release time, and dissemination support.

Letter of Collaboration: Dr. Sophia Patel, Director of the NSF AI Institute for Inclusive Learning, commits to collaborating through joint workshops, access to research materials, and participation in annual symposia.

Diversity and Inclusion Statement: This proposal centers inclusion at every level—recruiting underrepresented students, embedding ethics and equity in AI curriculum, and partnering with community organizations to expand participation.